

1.20 WIND MITIGATION

Objectives

- (a) To mitigate the effects of strong wind at street level.
- (b) To ensure wind does not preclude the functioning of the centre's key uses.
- (c) To encourage development to utilise the predominant breeze direction to inform an energy efficient design.

Controls

- (a) Buildings shall not create uncomfortable or unsafe wind conditions in the public domain that exceed the Acceptable Criteria for Environmental Wind Conditions.
- (b) Locate or design outdoor areas to ensure places with high wind level are avoided.
- (c) All applications for buildings over 5 storeys in height shall be accompanied with a wind environment statement, unless a wind tunnel study is required. For buildings over 9 storeys and for any other building which may be considered an exposed building, applications shall be accompanied by a wind tunnel study report (refer to *Annexure E1-1*).
- (d) Acceptable Criteria for Environmental Wind Conditions:

Area Classification	Limiting Weekly Maximum Gust-Equivalent Mean	Limiting Annual Maximum Gust
Outdoor dining areas, amphitheatres etc	3.5m/s	10 to 13m/s
Main retail centres such as Oxford St Mall, parks, communal recreational areas such as common swimming pool on the podium	5.5m/s	13m/s
Footpaths and other pedestrian access ways	7.5m/s	16m/s
Infrequently used laneways, easements, private balconies	10m/s	23m/s

Note: Gust-Equivalent Mean is defined as the maximum 3 second gust divided by a local Gust Factor for the local wind speed. It is recommended that the local gust factor be derived from the measured local turbulence intensity. If the mean wind speed happens to be greater than the Gust-Equivalent Mean then the Mean wind speed is to be adopted in place of the Gust-Equivalent Mean.

The Annual Maximum Gust wind speed criteria can be used as an alternative to the Gust-Equivalent Mean Criteria. If the Gust-Equivalent Mean criteria are being used then a check should also be made to ensure that all areas studied are within the Annual Maximum Gust wind speed of 23m/s.

When assessing the impact of a proposed development, no increase over the existing wind conditions is acceptable unless the increase over the existing conditions is such that the relevant criterion for that type of space is still satisfied.